

COMP1531

Software Engineering Fundamentals

Week 7

HTTP Server Coverage

We've already learned:

— — —

- `vitest run --coverage`

This works well for **unit tests** (**See Lecture 3.1 – Dynamic Verification (Slides 36–48)**)

Question ??

Can we use the same command to measure the coverage of an **Express server**?



Let's Try It: Measuring Code Coverage

— — —

Run the server

```
npm run start
```

Run the tests

```
npm run test -- --coverage
```

Expected?

Coverage of the Express server.

Reality?

Coverage report is **empty (0%)**

Output:

— — —

```
Test Files  1 passed (1)
Tests       3 passed (3)
Start at    02:19:31
Duration    269ms (transform 18ms, setup 0ms, import 76ms, tests 12ms, environment 0ms)
```

% Coverage report from v8

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	0	0	0	0	

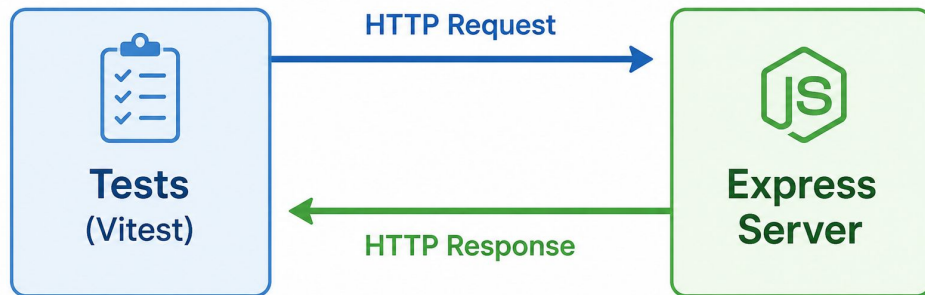
Why Doesn't vitest – coverage Work?

— — —

Our tests only send **HTTP requests**.

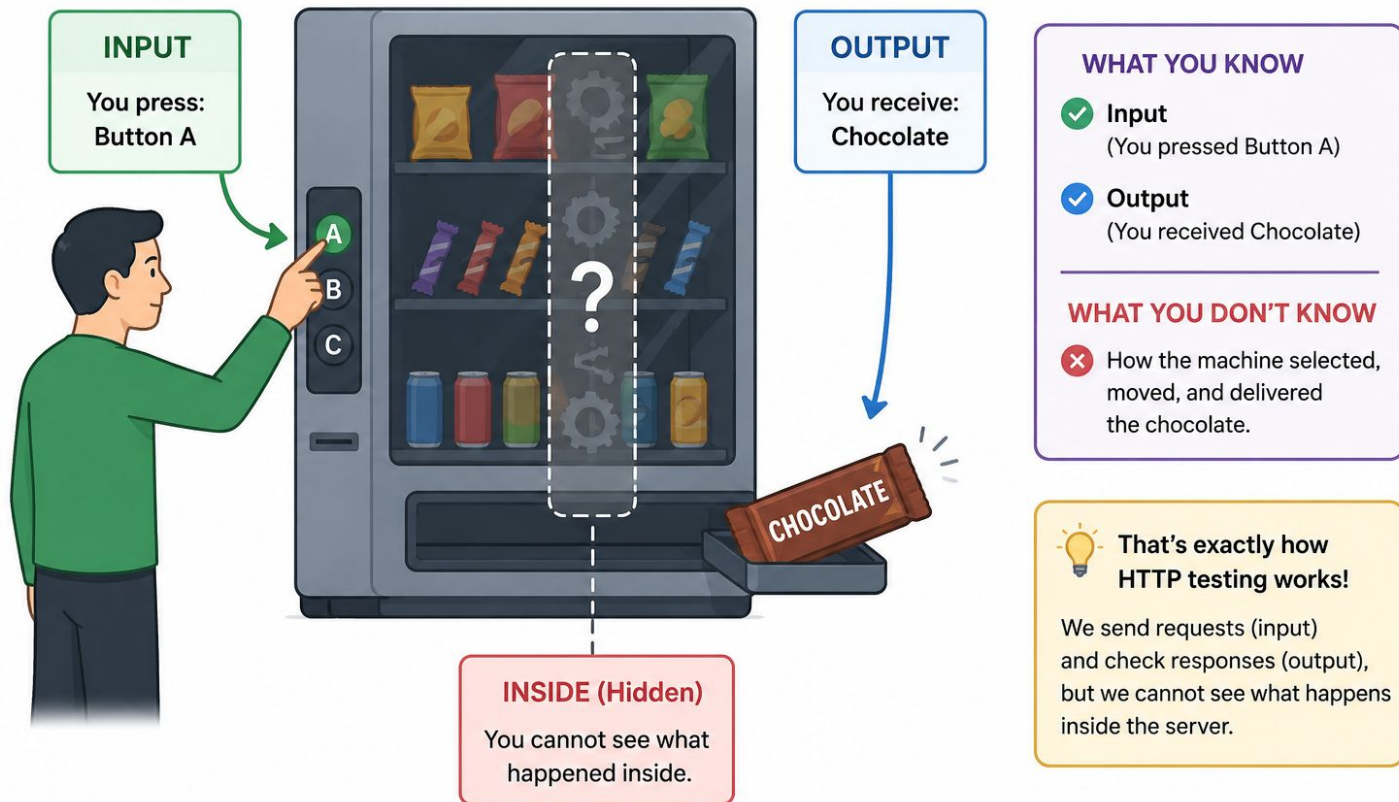
The tests only see:

- Request
- Response



They **cannot see the server code itself**.

Analogy: Testing a Vending Machine (Like HTTP Testing)



So How Do We Measure Server Coverage?

— — —

Instead of measuring the tests...

=> Measure the **server while it is running**.

We use a tool called :

C8

a **JavaScript code coverage tool** for **Node.js**. It measures **which lines of your program are actually executed** while your application is running.

c8 watches every line of code that executes.

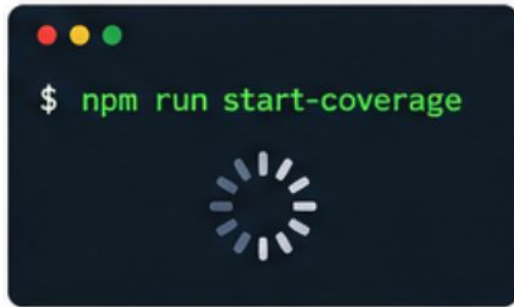
How do we proceed?

Step 1:

Start the server with **c8** coverage enabled.

```
npm run start-coverage
```

Instead of simply starting Express, the server now records every executed line.



How do we proceed?

Step 2:

Open another terminal.

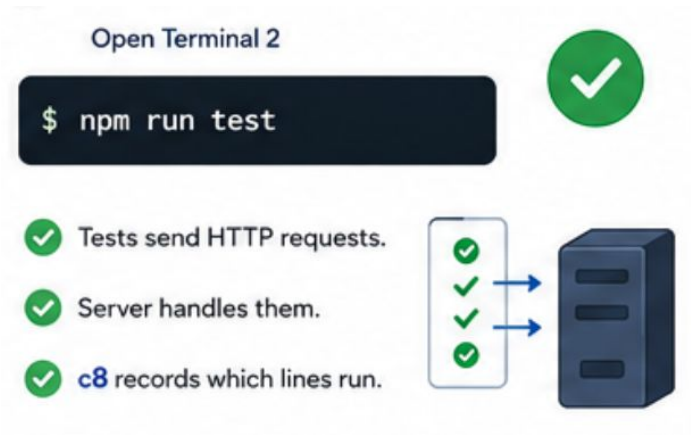
- Run your tests normally.

```
npm run test
```

The tests behave exactly the same.

The difference is:

The **server** is recording coverage.



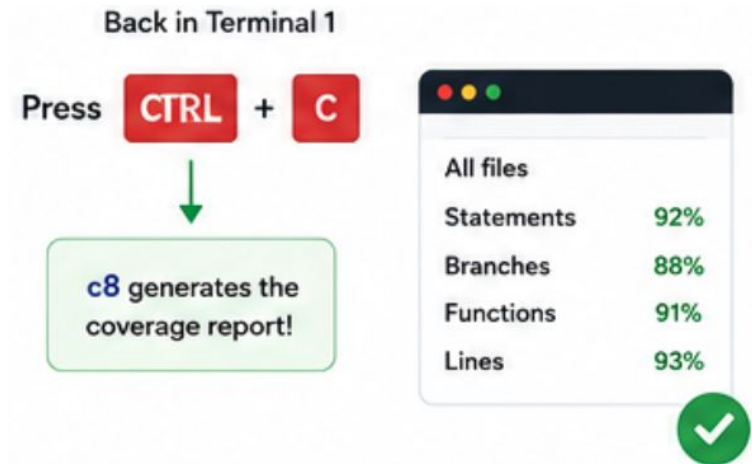
How do we proceed?

Step 3:

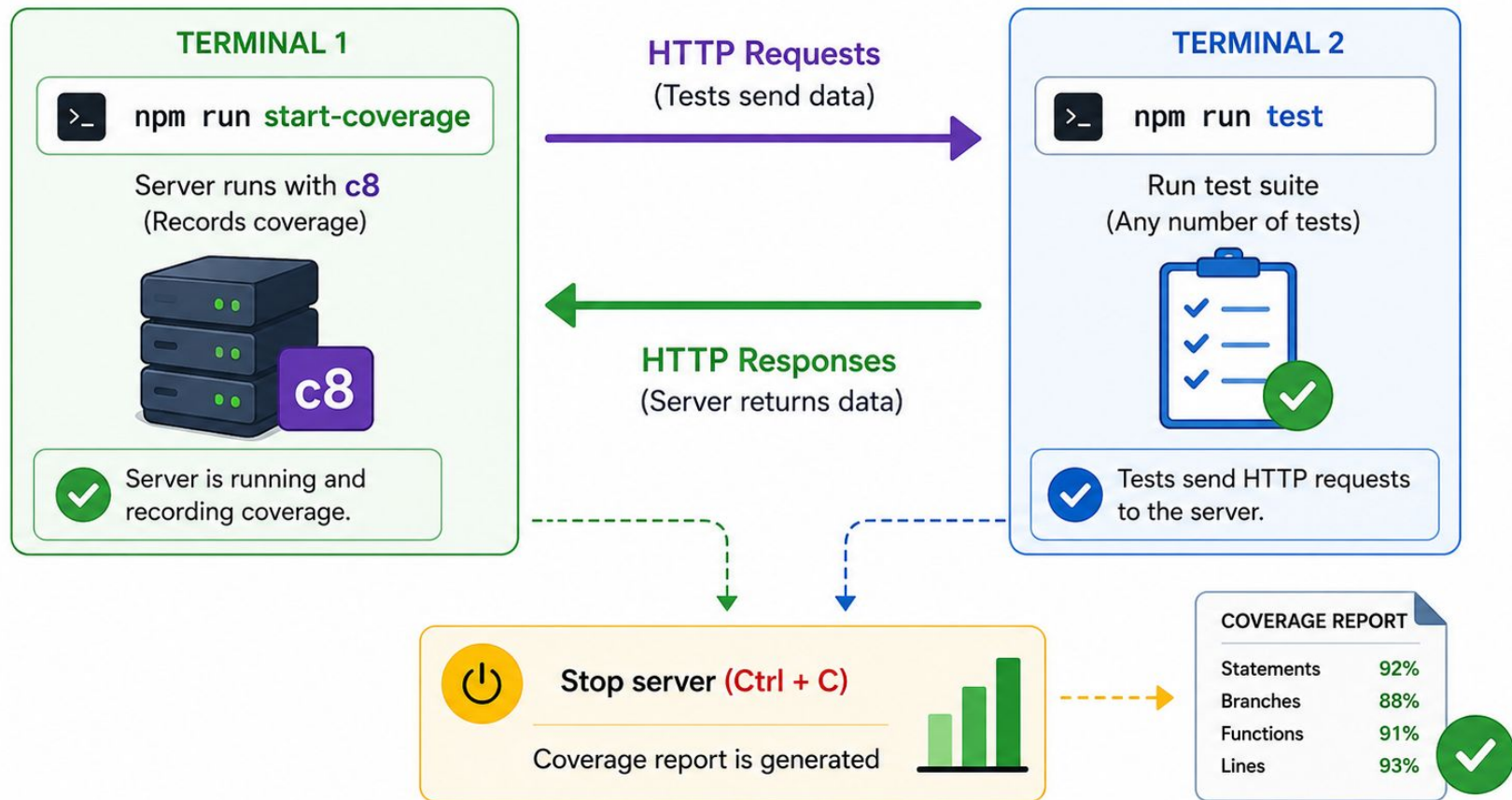
Stop the server

- Ctrl + C

Now c8 generates the coverage report.



Complete Workflow



Key Takeaways

- `vitest --coverage` measures **unit-tested code**
- HTTP tests only know the server's **inputs and outputs**
- They **cannot directly measure Express server code**
- Use **c8** to measure coverage while the server is running

★ Server coverage is a significant part of the **Iteration 2 automarking**, so always remember to run the server with `start-coverage` before testing.

Feedback

COMP1531 Feedback Survey

